Formal verification of two arithmetic-function ratio inequalities

Myeongjin Kang

25 October 2026

Abstract

We verify Conjectures 3 and 9 of S. I. Dimitrov's preprint *Lower bounds on expressions depending on the functions $\varphi(n)$, $\psi(n)$ and $\sigma(n)$, III*, arXiv:2606.12484v1. Both statements hold for every integer $n \geq 2$ and $k \geq 1$. Shivam Patel posted complete general arguments for these two statements on MathDB on 20 August 2026. This note credits those arguments and gives a derivative-free proof with a complete Lean certificate, including the arithmetic-function bridge. No novelty or equality-classification claim is made.

1 Statements and arithmetic functions

For $n \geq 1$, write $n = \prod_{p|n} p^{e_p}$ with $e_p \geq 1$. Let φ be Euler's totient, let $\sigma(n) = \sum_{d|n} d$, and define

$$\psi(n) = \prod_{p|n} p^{e_p - 1}(p+1).$$

Empty products have value one. This is precisely the definition of Dedekind psi in Section 1 of Dimitrov's preprint. It also satisfies

$$\psi(n) = n \prod_{p|n} (1 + 1/p),$$

since each factor on the right is $p^{e_p}(1 + 1/p) = p^{e_p - 1}(p+1)$.

Theorem 1 (Conjecture 3; MathDB 375256)**.** *For all integers $n \geq 2$ and $k \geq 1$,*

$$\frac{\varphi(n)^k}{\psi(n)^k + \sigma(n)^k} + \frac{\psi(n)^k}{\varphi(n)^k + \sigma(n)^k} + \frac{\sigma(n)^k}{\varphi(n)^k + \psi(n)^k} \geq \frac{4(n+1)^{2k} + (n^2-1)^k + (n-1)^{2k}}{2((n+1)^{2k} + (n^2-1)^k)}.$$

Theorem 2 (Conjecture 9; MathDB 375257)**.** *For all integers $n \geq 2$ and $k \geq 1$, put*

$$x = \varphi(n)^k, \quad y = \psi(n)^k, \quad z = \sigma(n)^k.$$

Then

$$\frac{x(y+z)}{y(x+z) + z(x+y)} + \frac{y(x+z)}{x(y+z) + z(x+y)} + \frac{z(x+y)}{x(y+z) + y(x+z)} \geq \frac{5(n-1)^{2k} + 5(n^2-1)^k + 2(n+1)^{2k}}{(3(n-1)^k + (n+1)^k)((n-1)^k + (n+1)^k)}.$$

Lemma 1 (Arithmetic bridge)**.** *For $n \geq 2$ and $k \geq 1$, with x, y, z as above and $a = (n-1)^k$, $b = (n+1)^k$, we have $0 < x \leq a < b \leq y \leq z$.*

Proof. The standard bounds $0 < \varphi(n) < n$ give $1 \leq \varphi(n) \leq n-1$. Each prime factor has positive exponent, and $p^{e-1}(p+1) = p^e + p^{e-1} > p^e > 0$. There is at least one prime factor when $n \geq 2$, so multiplying gives $\psi(n) > n$, hence $\psi(n) \geq n+1$ by integrality. The divisor-sum product formula gives

$$\sigma(n) = \prod_{p|n} (1 + p + \cdots + p^{e_p}).$$

Its last two summands dominate the corresponding psi factor; therefore $\sigma(n) \geq \psi(n)$. Raising these nonnegative bounds to the positive integer power k preserves weak inequalities and preserves $(n-1) < (n+1)$ strictly. □

2 Two real inequalities

Set $r = x/y$, $q = a/b$, and $t = z/y$. Lemma 1 implies $0 < r \leq q < 1$ and $t \geq 1$: in particular $x/y \leq a/y \leq a/b$. All denominators below are strictly positive. Define

$$S(r,t) = \frac{r}{1+t} + \frac{1}{r+t} + \frac{t}{r+1}.$$

The simple expression in Theorem 1 equals $S(r,t)$ by division of each numerator and denominator by y. Writing $v = t - 1 \geq 0$, direct common-denominator expansion gives

$$S(r,t) - S(r,1) = \frac{v\{2v^2 + (4 + r - r^2)v + r(1-r)(r+3)\}}{2(t+1)(r+t)(r+1)},$$
$$S(r,1) - S(q,1) = \frac{(q-r)\{4 - (1+r)(1+q)\}}{2(1+r)(1+q)}.$$

Both differences are nonnegative. Indeed $1-r \geq 0$, $4+r-r^2 = 4+r(1-r) \geq 0$, and $(1+r)(1+q) \leq 4$. Thus

$$S(r,t) \geq S(q,1) = \frac{a^2 + ab + 4b^2}{2b(a+b)}.$$

Since $ab = (n^2-1)^k$, $a^2 = (n-1)^{2k}$ and $b^2 = (n+1)^{2k}$, the last expression is the exact right-hand side of Theorem 1. This proves that theorem.

For the mixed expression, homogeneity of degree zero gives

$$G(r,t) = \frac{r(1+t)}{rt + r + 2t} + \frac{r+t}{2rt + r + t} + \frac{t(r+1)}{rt + 2r + t}.$$

Each mixed numerator and denominator was divided by y^2. Define the polynomial

$$\begin{aligned} P(r,v) = {} & (2r^4 + 4r^3 + 7r^2 + 8r + 3)v^2 \\ & + (4r^4 - 5r^3 + 8r^2 + 19r + 6)v \\ & + (1-r)(3r+1)(5r+3). \end{aligned}$$

The following two identities follow by expansion of common denominators:

$$\begin{aligned} G(r,t) - G(r,1) &= \frac{r(t-1)P(r,t-1)}{D(r,t)}, \\ D(r,t) &= (r+1)(3r+1)(rt+r+2t) \\ &\quad \cdot (rt+2r+t)(2rt+r+t), \\ G(r,1) - G(q,1) &= \frac{(q-r)(3+r+q-5rq)}{(r+1)(q+1)(3r+1)(3q+1)}. \end{aligned}$$

The apparent negative coefficient in P is harmless, because

$$4r^4 - 5r^3 + 8r^2 + 19r + 6 = 4r^4 + 5r^2(1-r) + 3r^2 + 19r + 6 \geq 0.$$

All other factors establishing $P(r,v) \geq 0$ are nonnegative. For the horizontal difference, $rq \leq r$ and $r \leq q \leq 1$ give

$$3 + r + q - 5rq \geq 3 - 4r + q \geq 3 - 3r \geq 0.$$

Consequently $G(r,t) \geq G(q,1)$, and elementary simplification gives

$$G(q,1) = \frac{5a^2 + 5ab + 2b^2}{(3a+b)(a+b)}.$$

The same power identities now give exactly the right-hand side of Theorem 2, proving it.

3 Formal certificate and provenance

The combined companion file `formal/ArithmeticRatios.lean` proves the identities, signs, arithmetic bridge, homogeneity, and the two exact final inequalities. The final declarations are `ArithmeticRatios.mathdb_375256` and `ArithmeticRatios.mathdb_375257`. They quantify over all natural n, k with $2 \leq n$ and $1 \leq k$, interpret all displayed quotients in $\mathbb{R}$, and use mathlib's actual `Nat.totient` and divisor-sum function. Subtraction on the right-hand sides is real subtraction, so there is no truncated-natural subtraction in those expressions.

The function `psi` is the integral product specified above. The theorem `psi_euler_product` formally proves its equality with $n\prod_{p\mid n}(1+1/p)$ when $n \neq 0$. The theorem `sigma_eq_sum_divisors` explicitly identifies `sigma` with $\sum_{d\mid n} d$. The code makes no conjecture-specific assumption and uses no `sorry`, newly postulated axiom, or `native_decide`. The replay script runs Lean 4.34.1 with warnings treated as errors; `formal/Checks.lean` reports axioms for every local declaration.

Dimitrov's v1 source has the exact statements under labels `Conjecture3` and `Conjecture9`. Shivam Patel posted both general proofs on 20 August 2026; versioned source locations and credited response identifiers are recorded in `provenance.json`. Patel's first proof uses derivatives; we replace those steps with explicit rational certificates. The mixed polynomial is his posted certificate. These are two distinct source questions sharing an arithmetic bridge, not a count of their helper lemmas. This is verification of prior posted proofs, with no claim of a first proof, worldwide priority, or an equality classification.

References

[1] S. I. Dimitrov, *Lower bounds on expressions depending on the functions $\varphi(n)$, $\psi(n)$ and $\sigma(n)$, III*, arXiv:2606.12484v1, 10 June 2026, https://arxiv.org/abs/2606.12484v1.

[2] Shivam Patel, posted proofs on MathDB questions 375256 and 375257, 20 August 2026. Source locations, timestamps, and solution identifiers are recorded in `provenance.json`; full third-party texts are retained only in the development checkout.